

electric flux

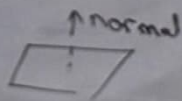
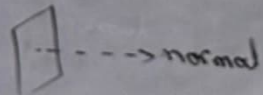
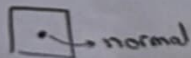
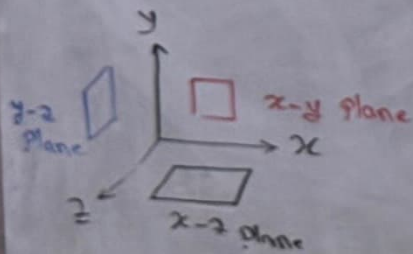
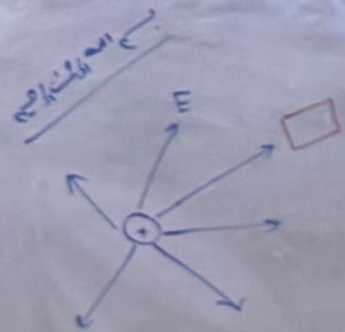
The number of E lines Penetrating some surface

$\phi_E \rightarrow$ electric flux ($N \cdot m^2/c$) $\rightarrow$ scalar quantity

$$\phi_E = \vec{E} \cdot \vec{A} \rightarrow \phi_E = EA \cos \theta$$

between E and Normal

* Normal: imaginary line ^{outward} perpendicular to the surface



electric flux

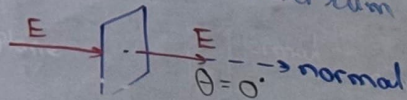
The number of E lines Penetrating some surface

$\Phi_E \rightarrow$ electric flux ($N \cdot m^2/c$) $\rightarrow$ scalar quantity

$$\Phi_E = \vec{E} \cdot \vec{A} \longrightarrow \Phi_E = EA \cos \theta \text{ or } \Phi_E = EA \sin \theta$$

$\swarrow$ between E and Normal $\swarrow$ between E and surface

① Maximum electric flux :

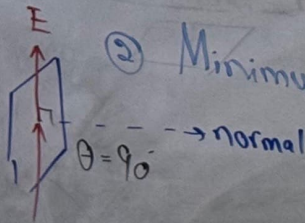


E is parallel to the Normal
* E is perpendicular to the surface

$$\theta = 0^\circ$$

$$\Phi_E = EA \text{ max.}$$

② Minimum electric flux :



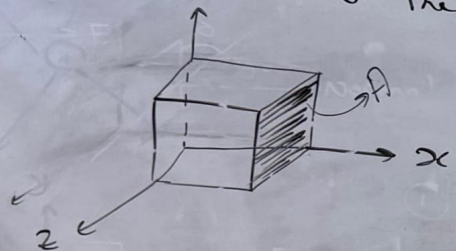
E is perpendicular to the Normal
* E is parallel to the surface

$$\theta = 90^\circ$$

$$\Phi_E = 0 \text{ min.}$$

مساحة المثلث

A cube of 60 cm side is placed in region in which $\vec{E} = 5\hat{i} - 4\hat{j} + 3\hat{k}$ N/C
Find the electric flux y-z Plane (shaded area)

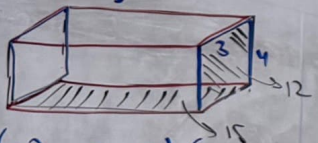


$$\Phi_E = \vec{E} \cdot \vec{A}$$

$$A = 0.6 \times 0.6 = 0.36 \text{ m}^2$$

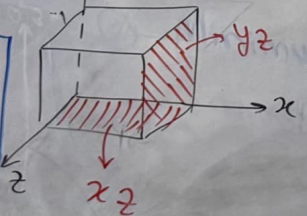
$$= (5\hat{i} - 4\hat{j} + 3\hat{k}) \cdot (0.36\hat{i}) = 1.8 \text{ N}\cdot\text{m}^2/\text{C}$$

Rectangular:



$$\begin{aligned} \Phi_E &= (5\hat{i} - 4\hat{j} + 3\hat{k}) \cdot (12\hat{i} + 3\hat{j}) \\ &= 5 \times 12 - 4 \times 3 = 0 \end{aligned}$$

Find the electric flux through x-z and y-z planes



$$A = 0.6 \times 0.6 = 0.36 \text{ m}^2$$

$$\Phi_E = (5\hat{i} - 4\hat{j} + 3\hat{k}) \cdot (0.36\hat{i} + 0.36\hat{j}) = 5 \times 0.36 - 4 \times 0.36 = 0.36 \text{ N}\cdot\text{m}^2/\text{C}$$